

PDF OCR and Complaint Extraction with Claude AI

Alexander Quispe
The World Bank, Caltech
aquisper@caltech.edu

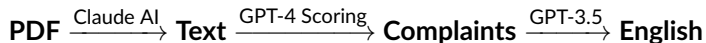
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Theoretical Framework: Processing Pipeline

Problem

- FIR documents in Hindi require automatic extraction and translation for legal analysis.

Solution: 3-Stage Pipeline



1. **OCR:** Extraction with Claude AI (better for Hindi).
2. **Identification:** Scoring by linguistic patterns.
3. **Translation:** GPT-3.5 preserving legal structure.

Result: Fully automated end-to-end system.

1. Extract text from PDFs with Claude

Objective

- Automate text extraction from PDF files using OCR with artificial intelligence.
- Process documents efficiently and in parallel.

Key Tools

- `anthropic`: Claude AI API for OCR.
- `pymupdf`: PDF file processing.
- `pillow`: Page to image conversion.
- `concurrent.futures`: Parallel processing.

Configuration

API Key and Model

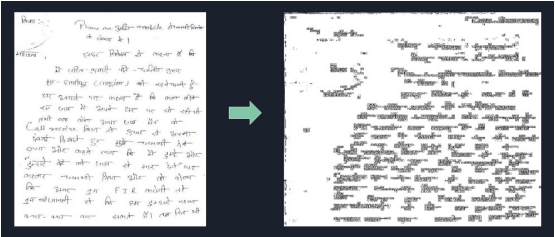
- The API key is securely stored in Google Drive.
- Uses the `claude-3-haiku-20240307` model from Anthropic.

Why Claude AI?

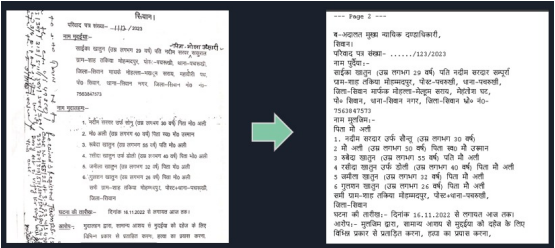
- Superior ability to recognize Hindi text.
- GPT showed difficulties with Devanagari script, generating empty transcriptions.
- Claude demonstrated higher accuracy in tests performed.

OCR Comparison

GPT Test



Claude Test



Processing Pipeline

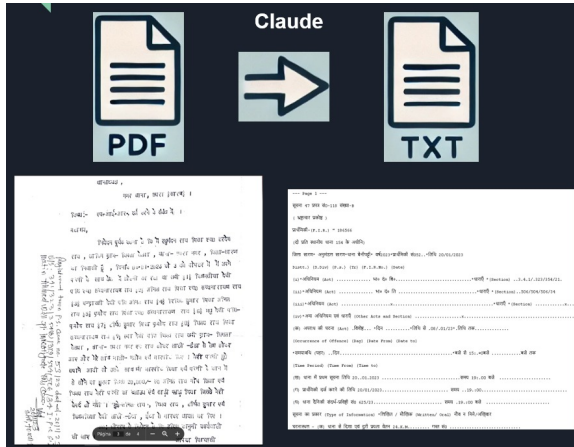
- **Step 1:** Mounts Google Drive and defines input/output folders.
- **Step 2:** Converts PDF pages to JPEG images.
- **Step 3:** Encodes images in base64.
- **Step 4:** Sends to Claude AI with retry system (3 attempts).
- **Step 5:** Processes 4 pages simultaneously with ThreadPoolExecutor.
- **Step 6:** Saves text in `.txt` files with page separators.

Results

System Outputs

- Generates structured .txt files.
- Avoids reprocessing existing documents.
- Fast processing through parallelism.

System Output



2. Identify complaint sections

Objective

- Automatically identify pages containing complaint or grievance sections in FIR documents.
- Extract and save relevant text from these sections for subsequent analysis.

Key Tools

- `re`: Regular expressions for pattern detection in Hindi text.
- `openai`: OpenAI API (version 0.28) as alternative processing.

Configuration

Paths and Files

- Input folder: `txt_transcribed/` (files extracted with Claude).
- Output folder: `txt_identified/` (identified complaint sections).
- OpenAI API key loaded from secure file.

Page Processing

- Splits documents using markers like “— Page X —”.
- Handles multiple page separator formats.
- Robust encoding error handling system (UTF-8, Latin-1, CP1252).

Identification Method

Complaint Indicators

- First-person pronouns (weight: 3 points).
- Formal petition language (weight: 5 points).
- Temporal references (weight: 3 points).
- Problems and requests (weight: 1-2 points).

Scoring System

- Each page receives a score based on indicator presence.
- Minimum threshold: 10 points for automatic identification.
- Fallback strategy uses typical pages by district.

Results

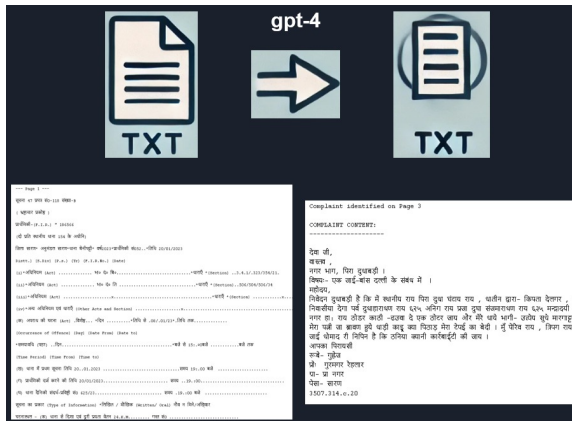
System Performance

- Successfully processed 12 FIR documents.
- Identified between 1-7 complaint pages per document.
- Saved extracted sections with preserved page markers.

Key Features

- Intelligent complaint start detection (skips headers).
- Automatic fallback for known districts (Saran, Siwan, Nalanda, etc.).
- High accuracy in identifying first-person narratives.

System Output



3. GPT-4 FIR complaint extractor

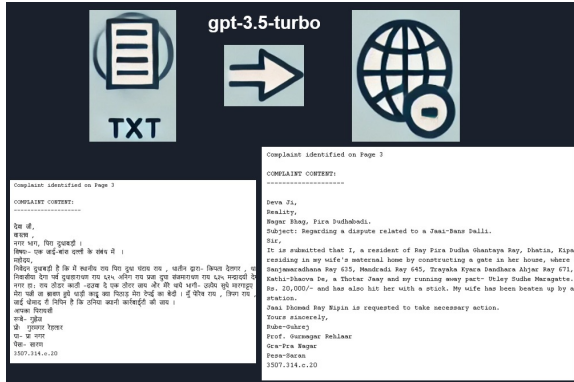
Objective

- Automatically translate identified complaint sections from Hindi to English.
- Preserve the structure, format, and original legal terms of the document.

Features

- Uses OpenAI API with gpt-3.5-turbo model.

System Output



Complete FIR Processing System



End-to-end automated pipeline:

OCR with Claude AI → Intelligent GPT-4 Identification → GPT-3.5 Translation

Efficient processing of Hindi legal documents